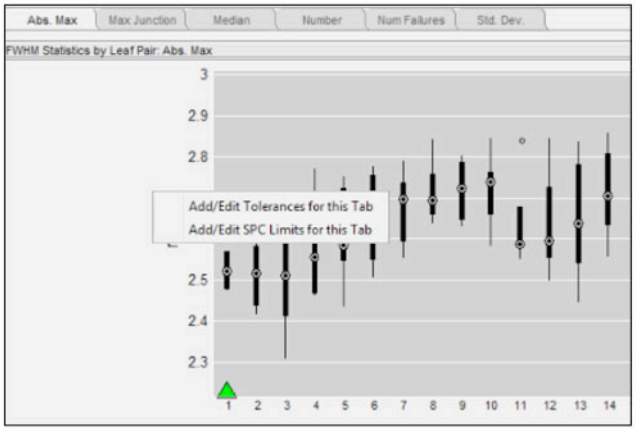
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Received: 14 June 2021 | Revised: 14 October 2021 | Accepted: 15 October 2021

DOI: 10.1002/mp.15312

RESEARCH ARTICLE

MEDICAL PHYSICS

Vision Transformer-based recognition of diabetic retinopathy grade

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Funding information
Key Projects of the National Natural Science Foundation of China, Grant/Award Number: 81630104; Major Project for Guang-Zhou Collaborative Innovation of Industry-University-Research, Grant/Award Number: 201704020196

Abstract

Background: In the domain of natural language processing, Transformers are recognized as state-of-the-art models, which opposing to typical convolutional neural networks (CNNs) do not rely on convolution layers. Instead, Transformers employ multi-head attention mechanisms as the main building block to capture long-range contextual relations between image pixels. Recently, CNNs dominated the deep learning solutions for diabetic retinopathy grade recognition. However, spurred by the advantages of Transformers, we propose a Transformer-based method that is appropriate for recognizing the grade of diabetic retinopathy.

Purpose: The purposes of this work are to demonstrate that (i) the pure attention mechanism is suitable for diabetic retinopathy grade recognition and (ii) Transformers can replace traditional CNNs for diabetic retinopathy grade recognition.

Methods: This paper proposes a Vision Transformer-based method to recognize the grade of diabetic retinopathy. Fundus images are subdivided into non-overlapping patches, which are then converted into sequences by flattening, and undergo a linear and positional embedding process to preserve positional information. Then, the generated sequence is input into several multi-head attention layers to generate the final representation. The first token sequence is input to a softmax classification layer to produce the recognition output in the classification stage.

Results: The dataset for training and testing employs fundus images of different resolutions, subdivided into patches. We challenge our method against current CNNs and extreme learning machines and achieve an appealing performance. Specifically, the suggested deep learning architecture attains an accuracy of 91.4%, specificity = 0.977 (95% confidence interval (CI) (0.951–1)), precision = 0.928 (95% CI (0.852–1)), sensitivity = 0.926 (95% CI (0.863–0.989)), quadratic weighted kappa score = 0.935, and area under curve (AUC) = 0.986.

Conclusion: Our comparative experiments against current methods conclude that our model is competitive and highlight that an attention mechanism based on a Vision Transformer model is promising for the diabetic retinopathy grade recognition task.

KEYWORDS
deep learning, diabetic retinopathy, multi-head attention, Vision Transformer

1 | INTRODUCTION

The International Diabetes Federation (IDF) collects data on people with diabetes from 110 countries, stating that in 2011, 366 million people had diabetes, which is expected to rise to 552 million by 2030, with the majority living in low- and middle-income countries.[1] Diabetic retinopathy (DR) is a disease that damages the retina due to diabetic complications, resulting in permanent eye damage and sometimes even vision loss.[2]

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Early detection and treatment of DR are essential to prevent vision loss.[3] In the deep learning era, several deep learning-based methods have been very successful in computer vision and image analysis tasks, significantly outperforming classic techniques. In the context of DR, Gulshan et al.[4,5] verify that deep learning algorithms are competitive with human experts in DR detection. Their method involves deep convolutional neural networks (CNNs) challenged on two DR datasets annotated by more than seven ophthalmologists. Keel et al.[6] visualize a deep CNN's output to highlight lesions in DR and verify the model's effectiveness by testing it on a manually annotated dataset by human experts. Among the various deep learning methods utilized for DR detection, the most widely used are CNN-based approaches. CNNs utilize convolutional kernels for scanning images with small perceptual fields and extracting image features through local connectivity and weight sharing, significantly reducing the parameter cardinality compared to fully connected networks. Many researchers have proposed different methods for DR detection. For example, Roy et al.[3] combine CNNs with dictionary-based approaches, incorporating pathology-specific image representation into the learning framework. This strategy affords improved DR severity classification, achieving a quadratic kappa score of $k^2 = 0.86$. Accordingly, Shankar et al.[7] suggest a deep neural network-based moth search optimization (DNN-MSO) algorithm appropriate for DR detection and classification. The DNN-MSO pipeline involves the stages of preprocessing, segmentation, feature extraction, and classification. Its maximum accuracy is 99.12%, sensitivity 97.91%, and specificity 99.47%. Bodapati et al.[8] develop a composite deep neural network structure incorporating a gated attention mechanism. This network facilitates feature descriptors obtained from multiple pre-trained deep CNNs to represent fundus retinal color images. The accuracy of this composite neural network is 82.54%, with a kappa score of 0.79. Takahashi et al.[9] suggest an improved GoogLeNet having a reduced batch size of four that detects various DR stages with 81% accuracy and 0.74 kappa score. Li et al.[10] employ a modified U-Net to detect secretions. The model is trained on the eophtha dataset achieving 96% area under curve (AUC) on DIARETDB1. Combining local and global CNN architectures has also been proposed.[11] Indeed, a local CNN comprising 10 layers employed for lesion detection and a global CNN with 26 layers for DR classification detection manages a final AUC of 0.9687, specificity of 89.80%, and sensitivity of 95.90%.

Alternatively to CNN-based approaches, a deep learning method entitled Transformer has also been proposed achieving great success on vision tasks.[12] The Transformer relies on a powerful attention mechanism, which focuses on certain parts of the input image to obtain more efficient results. It currently is the norm for natural language processing[13] and is also utilized in machine translation,[14] language modeling,[15] and speech recognition.[16]

Spurred by the great success of Transformer in natural language processing (NLP), researchers extend Transformer in image processing tasks. Nevertheless, this is a very challenging task. Compared to language sequences, images are matrices with spatial structures containing many pixels, forcing the attention mechanism to estimate each pixel-pair interplay and imposing a very high computational cost. To solve this drawback, Wang et al.[17] generate powerful features by selectively focusing on critical parts or locations of an image and then processing them sequentially. Wu et al.[18] cascade a CNN with a Transformer, where the CNN extracts the feature maps, which are then input to a stacked Vision Transformer (ViT) to process visual tokens and compute the classification output. Chen et al.[19] apply the Transformer model to raw low-resolution images and reshape them into text-like long pixel sequences.

Nevertheless, none of these approaches sufficiently affects the CNNs' dominance in vision tasks. Recently, Dosovitskiy et al.[13] subdivide an image into non-overlapping patches and input a sequence of these patch embeddings into a Transformer, with the latter exploiting a pure multi-head self-attentive mechanism that has achieved highly competitive success on the ImageNet dataset. In image classification tasks, where convolutional network architectures dominate, the ViT proposed by Dosovitskiy et al.[13] demonstrates that the dependence on CNN is unnecessary and that a pure Transformer applied directly to a sequence of image patches can classify images very well.

Transformers have already been studied on medical imagery tasks such as DR classification. A lesion-aware Transformer (LAT) proposed by Sun et al.[20] achieves DR grading and lesion discovery through an encoding–decoding architecture. The Encoder relies on pixel relations, while the decoder on a lesion filter. However, this architecture contains both the encoding and decoding parts of the original Transformer, while ViT uses only its encoding part. Dalmaz et al.[21] develop a generative adversarial model for medical image synthesis named ResViT by combining the local localization power of convolutional operators with the context sensitivity of ViT. ResViT outperforms state-of-the-art CNN-based and transformer architectures in several medical imaging tasks. Kamran et al.[22] propose a ViT-based generative adversarial network (VTGAN) that simultaneously produces retinal vascular structures from fundus images and predicts retinal degeneration. However, the model's prediction capability is binary, and it does afford grade recognition of the disease. In a recent study, CNNs and ViT were used for the diagnosis of pneumonia through transfer learning.[23] In the comparison between the ViT model and the seven CNNs models, ViT achieved the highest Recall, and for several other evaluation

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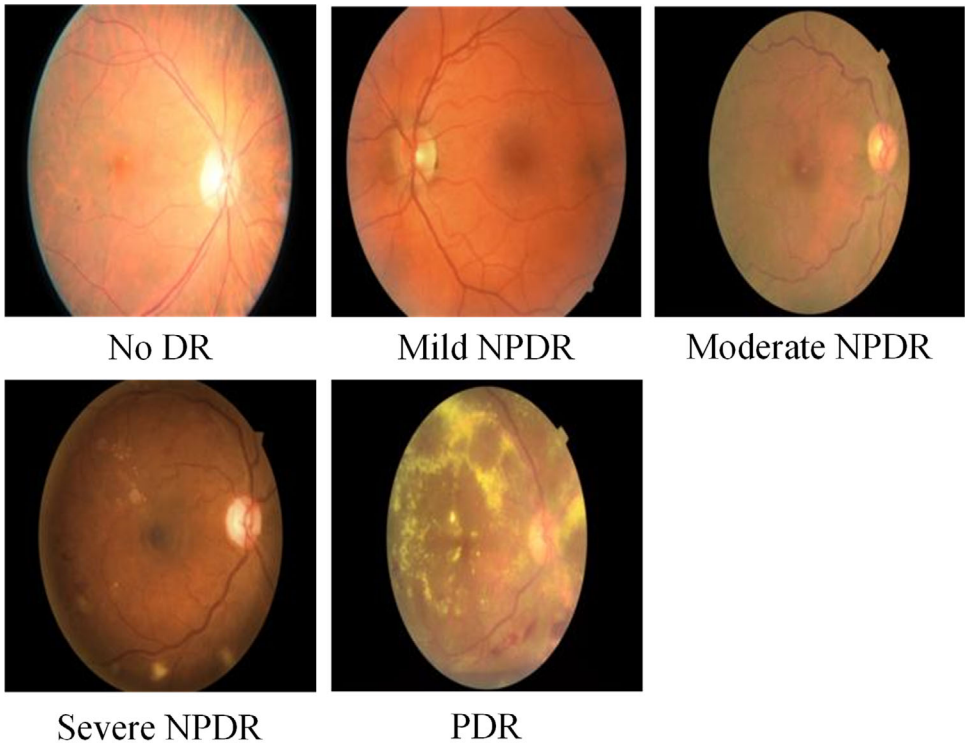


FIGURE 1 Five grades diabetic retinopathy (DR) fundus images. There are two stages of DR: early and advanced. In the former stage, new blood vessels do not develop (proliferate), often referred to as non-proliferative diabetic retinopathy (NPDR). The vessel walls within the retina weaken due to NPDR, and narrower bulges (microaneurysms) protrude from the surface of the narrower vessels, often dripping fluid and blood into the eye. Additionally, retinal vessels begin to dilate and become irregular in diameter. The NPDR disease progresses from mild to severe as more vessels become blocked and are divided into three stages: mild, moderate, and severe.[24] Advanced DR, also named proliferative diabetic retinopathy (PDR), is the condition where damaged blood vessels leak the clear, gelatinous fluid that fills the eye's center (vitreous humor) and leads to the development of abnormal blood vessels in the retina. The new growth of blood vessels interrupts the normal fluid flow, increasing the eye pressure damaging the optic nerve that transmits images from the eye to the brain, ultimately leading to glaucoma

metrics, ViT also showed a considerable competitive advantage. Notably, ViT has a much larger number of parameters than the other seven CNNs models, which indicates the great potential of ViT for medical image processing tasks.

This paper applies the ViT architecture for DR grade recognition and, specifically, classifying fundus images into five grades (Figure 1): no DR, mild non-proliferative diabetic retinopathy (NPDR), moderate NPDR, severe NPDR, and proliferative diabetic retinopathy (PDR). To perform ViT for DR grade recognition, we adopt the design of Dosovitskiy et al.[13] and subdivide the fundus imagery into non-overlapping patches, which are converted into sequences through flattening and embedding. Additionally, to preserve the position information of these patches, their position embedding is added. Next, the generated patch sequence is input into several multi-head attention layers to generate the final representation. The first token sequence is input to the softmax classification layer during the classification stage to produce the recognition output.

The remainder of this paper is organized as follows: Section 2 describes the ViT-based approach. Section 3 introduces the relevant datasets and experiments, while Section 4 analyzes and discusses the results. Finally, Section 5 concludes this work.

2 | MATERIALS AND METHODS

A ViT uses only the encoder block of the original Transformer[25] and connects the classification header to the final output of the encoding. The ViT structure is illustrated in Figure 2.

We subdivide the fundus image into fixed-size, non-overlapping patches and embed these patches linearly. Then, we add positional embedding and input the generated vectors to the Encoder. Regarding the latter, we add a classification head to its final output to facilitate the classification task. Throughout the process, the encoder block iterates L times.

2.1 | Vision Transformer

The original version of the Transformer exploits as input a one-dimensional (1D) sequence token embedding, and therefore some image preprocessing is required

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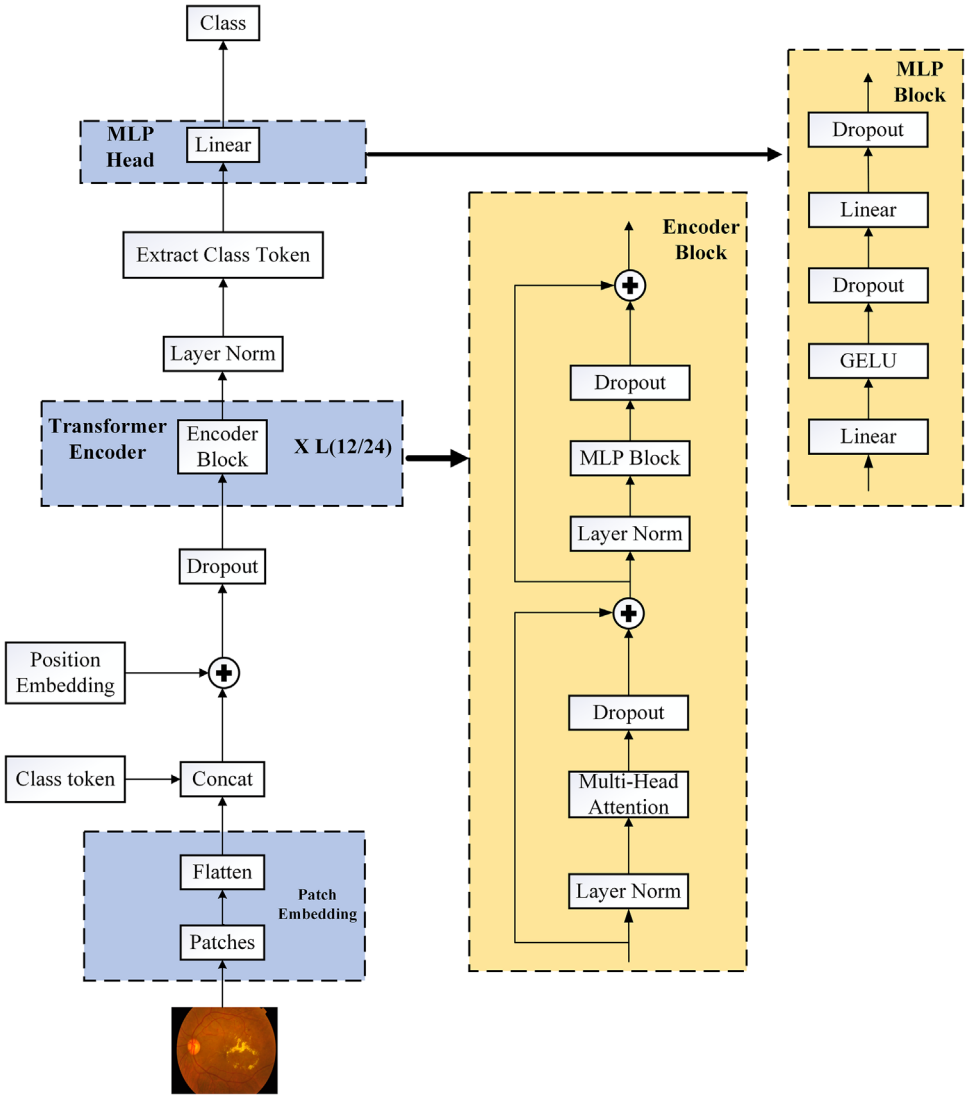


FIGURE 2 Vision Transformer (ViT) structure (from left to right: overall ViT architecture, encoder block internal structure, multi-layer perceptron internal structure). Notably, the input fundus images are transformed into a series of tokens, and a token dedicated to classification is placed before them, and this token for classification is trainable. Similarly, the positional embedding is trainable, and it performs element-wise summation with the previous tokens, so they have the same shape

to process two-dimensional (2D) images. Let $S = \{X_i, y_i\}_{i=1}^{k}$ denotes k fundus images, where X_i is a fundus image and y_i is the label corresponding to X_i with $y_i \in [0, 1, 2, 3, 4]$. The aim of ViT is learning the mapping from the patch sequence to the corresponding label y_i.

A fundus image $X \in \mathbb{R}^{H\times W\times C}$ is subdivided into N non-overlapping 2D patches $x_p \in \mathbb{R}^{P\times P\times C}$, where $(H \times W)$ is the original fundus image size, C is the number of channels, $(P \times P)$ is the resolution of each patch, and $N = !!!!!(H \times W)/P^2$. In this work, we set $H = W = 224$ or 384 and $P = 16$ or 32.

Before we input the patch sequence to the Encoder, each patch is linearly projected into a fixed D-dimensional vector utilizing a learnable embedding matrix E. The latter is added to the head of the embedded patch sequence, which is used as the image representation y in ViT. To preserve the spatial information of the patches in the original fundus image, the position information (Figure 3) is embedded into the patch embedding through encoding, with the entire process presented in Equation (1). Equation (2) is the position information encoding, where pos is the position of the patches and i is the dimensional position on the D-dimensional vector.

$$z_0 = \left[x_{\text{class}}; x_P^1 E; x_P^2 E; \cdots; x_P^N E\right] + E_{\text{pos}}, E \in \mathbb{R}^{(P^2C)\times D},$$

$$E_{\text{pos}} \in \mathbb{R}^{(N+1)\times D} E_{(\text{pos},2i)} = \sin\left(\frac{\text{pos}}{10000^{2i/D}}\right) \quad (1)$$

$$E_{(\text{pos},2i+1)} = \cos\left(\frac{\text{pos}}{10000^{2i/D}}\right) \quad (2)$$

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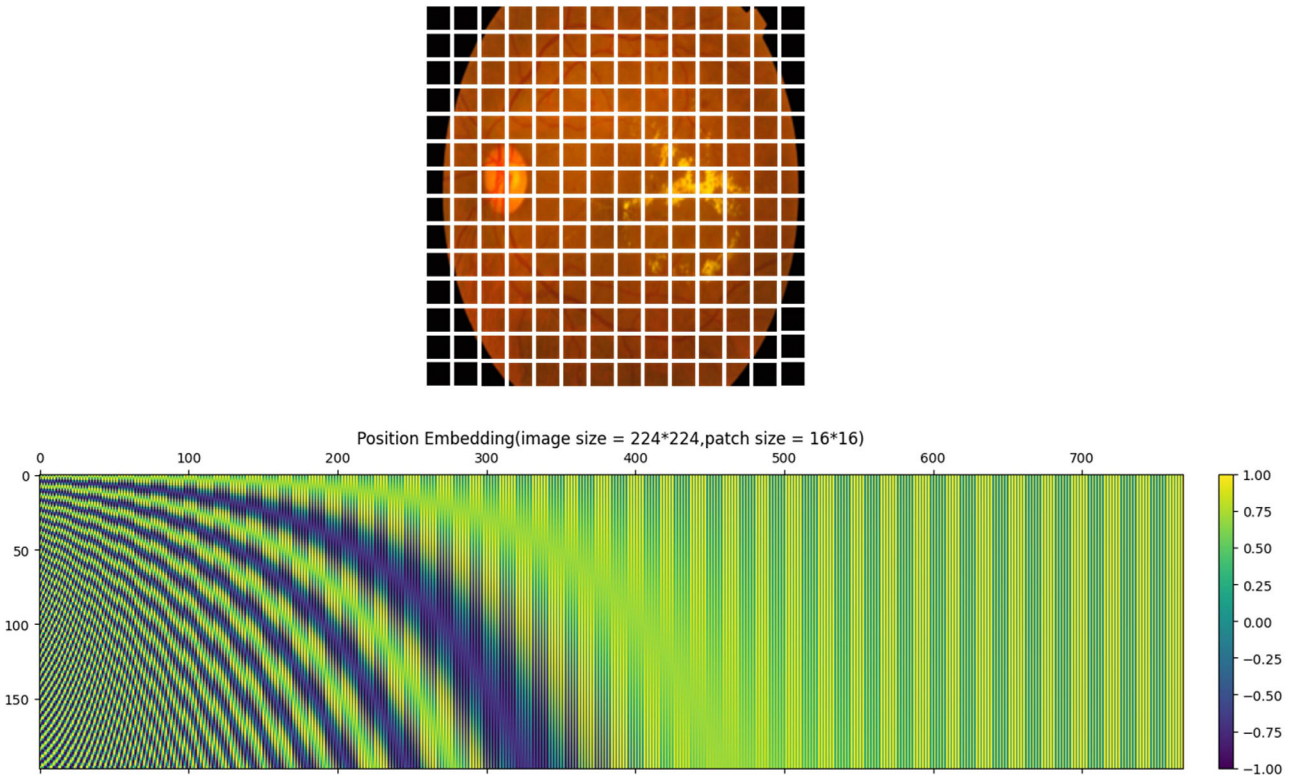


FIGURE 3 Position information visualization. Top image: a 224 × 224 fundus image is subdivided into 16 × 16 patches by a convolution operation with a stride size of 16 × 16, and a convolution kernel size of 16 × 16. Bottom image: the position information dimension of this fundus image is 197 × 768 calculated by Equation (2)

Then Z_0 is input to ViT's Encoder, comprising L identical layers containing a multi-head self-attention (MSA) and an multi-layer perceptron (MLP) module. The latter module consists of two fully connected layers with a gaussian error linear units (GeLu) activation layer in between. Both MSA and MLP use residual connections and are preceded by a layer normalization (LN) process.

$$Z_l' = \mathrm{MSA}\left[\mathrm{LN}\left(Z_{l-1}\right)\right] + Z_{l-1}, \quad l = 1, 2, \ldots L \tag{3}$$

$$Z_l = \mathrm{MLP}\left[\mathrm{LN}\left(Z_l\prime\right)\right] + Z_l', \quad l = 1, 2, \ldots L \tag{4}$$

$$y = \mathrm{LN}\left(Z_L^0\right) \tag{5}$$

$$\mathrm{GeLu} = 0.5x\left\{1 + \tanh\left[\sqrt{\frac{2}{\pi}}\left(x + 0.044715x^3\right)\right]\right\} \tag{6}$$

2.2 | Self-attention

Attention mechanisms have become an essential component in various image processing and NLP tasks and have also been combined with CNNs for specific tasks.[26,27] An attention module is represented by the attention weight, a weighted sum of all values in the sequence $Z \in \mathbb{R}^{N\times D}$. Self-attention (SA) learns the attention weight by computing the scaled dot product between query–key–value (shown in Figure 4a), and for each element in the input sequence, it generates its own three values: Q (query), K (key), and V (value) through a learnable matrix $U_{QKV} \in \mathbb{R}^{D\times 3D_k}$ (Equation (7)). The correlation between the two elements is determined by calculating the dot product of one of these elements' Q-vector with the K-vector of the other element. The dot product result is scaled, and input into the softmax layer to form the patches' relative importance value within the sequence (Equation (8)). Finally, the SA module is given by Equation (9).

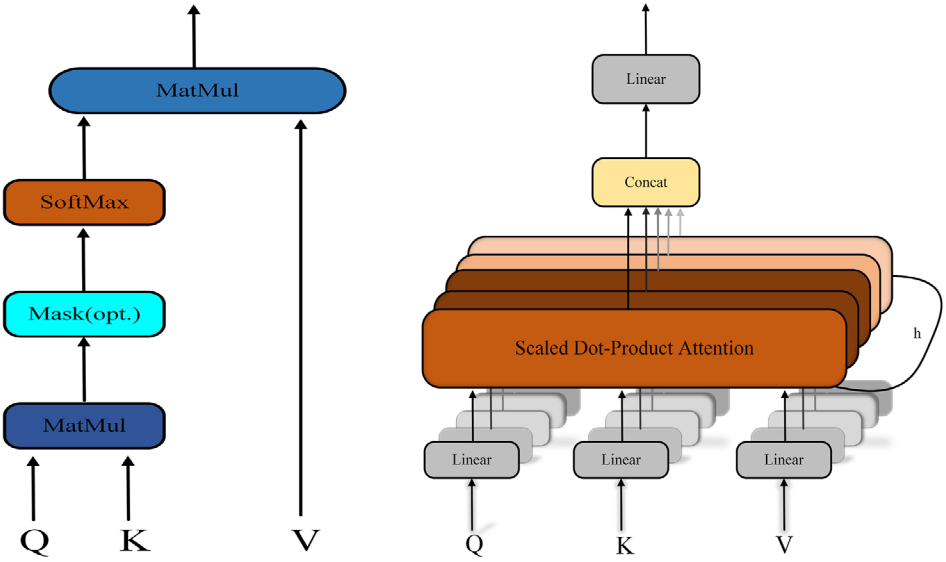
$$[Q, K, V] = ZU_{QKV}, Z\epsilon\mathbb{R}^{N\times D}, U \in \mathbb{R}^{D\times 3D_k} \tag{7}$$

$$A = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{D_k}}\right), A \in \mathbb{R}^{N\times N} \tag{8}$$

$$\mathrm{SA}(Z) = A \cdot V \tag{9}$$

where D_k is the dimension of K.

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(a). Self-attention (b). Multihead self-attention

FIGURE 4 Attentional mechanisms. Multi-headed self-attention module (b) is the integration of multiple self-attention modules (a)

2.3 | Multi-head self-attention

An MSA is an extension of SA involving h SA operations in parallel, called h "heads", which linearly project the cascaded output of all results to the desired dimension through a learnable projection matrix E_{MSA}. The operation diagram of the MSA is illustrated in Figure 4b and is expressed as:

$$\mathrm{MSA}(Z) = \mathrm{concat}\left[\mathrm{SA}_1(Z); \mathrm{SA}_2(Z); \cdots; \mathrm{SA}_h(Z)\right] E_{\mathrm{MSA}},$$
$$E_{\mathrm{MSA}} \in \mathbb{R}^{hD_k \times D} \tag{10}$$

2.4 | Model structure

The proposed ViT architecture aims to recognize five different DR grades: no DR, mild NPDR, moderate NPDR, severe NPDR, and PDR. Thus, the fundus image is subdivided into (16 × 16) or (32 × 32) pixels patches that are non-overlapping, and then each patch is linearly projected into a fixed D-dimensional vector using a learnable embedding matrix E. Next, a learnable embedding is added to the head of the embedded patches sequence as a classification representation to preserve the patches' spatial information in the original fundus image. The position information is embedded into the patches embedding through the encoding process. Finally, the generated embedding is input into the Encoder, and after utilizing an MSA operation and an L-layer Encoder iteration, the first token is output as the final recognition result in the output sequence. The entire classification process is illustrated in Figure 5, involving a fixed fundus image resolution of (224 × 224) and (384 × 384), respectively. The main ViT training steps are presented in Algorithm 1. It is worth noting that the fine-tuning code is available at https://github.com/lukemelas/PyTorch-Pretrained-ViT, the pre-trained models are available at https://github.com/lukemelas/PyTorch-Pretrained-ViT/releases.

Algorithm 1: Vision Transformer (ViT)

Input: Fundus image dataset $S = \{X_i, y_i\}_{i=1}^{k} y_i \in [0, 1, 2, 3, 4]$
Output: Predicted labels
Set the number of iterations epoch, loss function, training batch_size, optimizer, learning rate lr.
1. For epoch = 1 : epoch:
2. for batch = 1 : k/batch_size:
3. Subdivision of the original fundus image into non-overlapping patches.
4. Linear projection of each patch into a fixed D-dimensional vector using the learnable embedding matrix E.
5. Add a learnable embedding to the head of the embedded patches sequence as a classification representation.
6. The location information is embedded in the patches embedding by encoding.
7. The generated embedding is fed to the Encoder.
8. MSA (Equation (3)) operation and L-layer encoder iteration.
9. Calculating losses
10. Back-propagation loss
11. Update model parameters
12. Output model prediction results

2.5 | Data augmentation strategies

Data augmentation increases the number and diversity of training datasets while reducing the risk of model overfitting. It generates additional data samples of the

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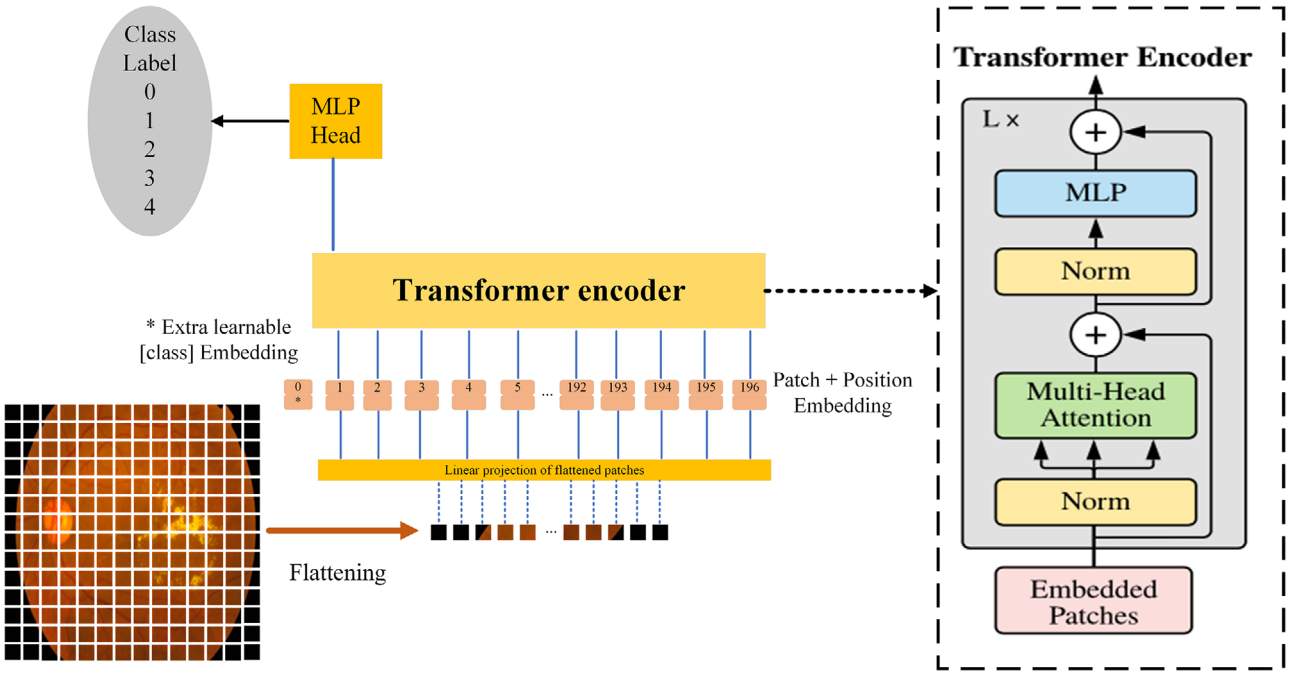


FIGURE 5 Vision Transformer (ViT) model structure for diabetic retinopathy (DR) grade recognition. The structure is consistent with Figure 2

same class label by applying simple geometric transformations on the original data, including rotation, flipping, cropping, scaling, panning, or their combination. Alternative data augmentation strategies consider adjusting the brightness, contrast, and color saturation of an image. Data augmentation is a highly effective solution, especially when large datasets are unavailable.[32] Other data augmentation strategies involve a neural style transfer that augments data by transferring the style of one image to another image while preserving the original semantic content.[33] Alternatively, the generative model-based approach generative adversarial network GAN[34] generates data similar to the original sample by learning the data distribution.

2.6 | Model evaluation metrics

There are several metrics that are widely used in image recognition tasks: accuracy, precision, recall (sensitivity), specificity, and F1 score. Each metric is calculated as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{11}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{12}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{13}$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{14}$$

$$\text{F1 score} = \frac{2\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{15}$$

Due to the between-classes image imbalance, this study also evaluates the number of misclassified images and penalizes the accuracy score accordingly. Thus, we utilize the quadratic weighted kappa score metric to measure the consistency between the correctly and misclassified ratings. For the dataset employed, each image is characterized by a tuple (e_h, e_t) that corresponds to the ratings of a human physician rater H and the proposed model rate T. To calculate the quadratic weighted kappa, an $(N \times N)$ histogram matrix O is constructed, with $O_{i,j}$ corresponding to the number of fundus images i and j rated from H and T, respectively. Then, an N-by-N weight matrix w is calculated from the rating differences between the raters as follows.

$$w_{i,j} = \frac{(i-j)^2}{(N-1)^2} \tag{16}$$

Assuming no correlation between ratings, the histogram matrix E of the expected ratings is calculated, which is the outer product between the histogram vectors of each rater's ratings. By normalizing E and O, both have the same sum. The quadratic weighted kappa calculated

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TABLE 1 Class distribution in original dataset

Class	Name	Number of images	Percentage
0	No DR	25 810	73.48
1	Mild DR	2443	6.95
2	Moderate DR	5292	15.07
3	Severe DR	873	2.48
4	PDR	708	2.02

Abbreviations: DR, diabetic retinopathy; PDR, proliferative diabetic retinopathy.

TABLE 2 Class distribution after augmentation

Class	Name	Number of images	Percentage
0	No DR	25 810	24.56
1	Mild DR	24 387	23.20
2	Moderate DR	26 460	25.18
3	Severe DR	15 708	14.94
4	PDR	12 736	12.12

Note: Data augmentation strategies include panning and random rotation.
Abbreviations: DR, diabetic retinopathy; PDR, proliferative diabetic retinopathy.

from these three matrices is:

$$k = 1 - \frac{\sum_{i,j} w_{i,j} \times O_{i,j}}{\sum_{i,j} w_{i,j} \times E_{i,j}} \tag{17}$$

3 | RESULTS

3.1 | Related datasets

This work considers the Diabetic Retinopathy Detection,[35] a public dataset from Kaggle. Table 1 presents the data distribution per class, demonstrating a severe imbalance imposing overfitting of most classes, that is, classes 0 and 2, and underfitting for the remaining classes. We augment the data by panning and randomly rotating the available training imagery to solve this problem. The panning scales are 10, 15, 20, 25 pixels, and the random rotation angles are 90°, 180°, 270°.

Additionally, the dataset employed involves fundus images acquired by different devices and lighting conditions. To eliminate the latter's effect, we perform histogram equalization on the original images and then apply our data augmentation strategy. The distribution of the augmented data is presented in Table 2, and despite being still unbalanced, it is better balanced than the original dataset. We divide the dataset into a training set and a test set according to an 8:2 ratio. The corresponding fundus images per class used for training and testing are presented in Table 3. In our trials, we also investigate the interplay between classification performance and various image sizes, that is, (224 × 224) and (284 × 384) pixels, and of various patch sizes, that is, (16 × 16) and (32 × 32) pixels.

TABLE 3 Data distribution for training and testing

Class	Training set	Test set
0	20 648	5162
1	19 510	4877
2	21 168	5292
3	12 566	3142
4	10 189	2547

TABLE 4 Parameter statistics for Vision Transformer (ViT)-B and ViT-L

Model	Number of layers	Hidden dimension	MLP size	Number of heads	Number of participants
ViT-B	12	768	3072	12	86M
ViT-L	24	1024	4096	16	307M

3.2 | Experiments

All experiments are implemented on an Intel(R) Core(TM) i9-10920X CPU @ 3.50 GHz computer with 32.0 GB RAM, utilizing an NVIDIA GeForce GTX 3080 10G graphical processing unit (GPU). The code is implemented in Pytorch 1.8.1, an open-source deep neural network library written in Python, and we used Python version 3.7.9.

Both fundus image resolutions, that is, (224 × 224) and (384 × 384) pixels, are employed to train the ViT configurations of ViT-B and ViT-L, with their parameter setup presented in Table 4. Accordingly, each model is trained with two patch size variants (16 × 16) and (32 × 32), respectively. For the (224 × 224) pixel fundus image, the model is pre-trained on the ImageNet-21k dataset, denoted as vit_224_B_16, vit_224_B_32, and vit_224_L_32, respectively. Accordingly, for the (384 × 384) pixel fundus image, the model is pre-trained on the ImageNet-21k dataset and fine-tuned on the ImageNet-1k dataset, denoted as vit_384_B_16, vit_384_B_32, vit_384_L_16, and vit_384_L_32, respectively.

Model vit_384_L_16 requires 37 h for training, and the remaining six models, 9 h per model. All experiments employ the CrossEntropyLoss function as the loss function due to the unbalanced data distribution. Loss weights are set to [1.0,1.0,1.0,2.0,2.0] according to the data distribution so that the model is fully trained. Adam is the optimizer with a learning rate of 2e-4. In addition to the data augmentation strategy, we also use the early stopping strategy to prevent overfitting.

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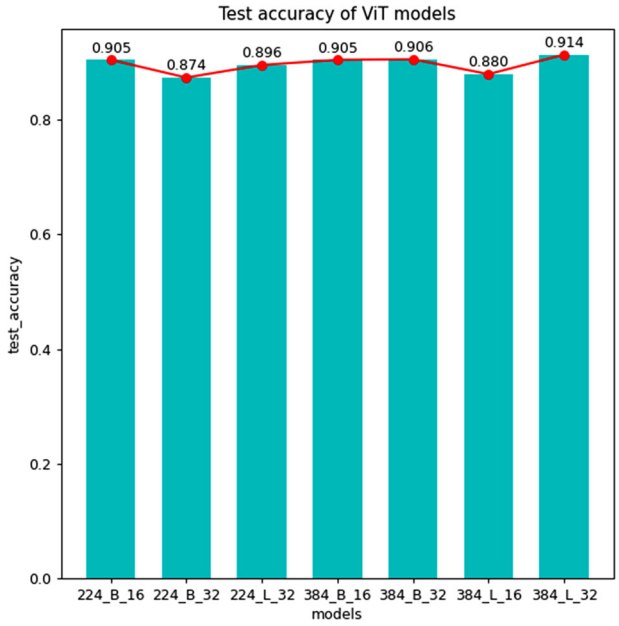


FIGURE 6 The test accuracy of each model

Once training completes, we test each model's accuracy on the test dataset, with the corresponding results presented in Figure 6. To comprehensively evaluate the model's performance, we calculate the model's precision, recall (sensitivity), specificity, and F1 score.

Figure 6 highlights that 224_B_32 has the lowest test accuracy attaining 0.874, while 384_L_32 achieves the highest accuracy of 0.914, followed by 384_B_32. Regarding the two image resolutions and while preserving the model's layers and number of heads, the test accuracy utilizing (384 × 384) pixel imagery is higher than that of (224 × 224) pixels.

Table 5 presents the performance metrics per DR grade and model, with the metric attaining the highest performance highlighted in boldface. Figure 6 highlights that the 384_L_32 and 384_B_32 models achieve the highest testing accuracy, that is, 0.914 and 0.906, respectively. From Table 5, it is evident that these two models still attain the highest performance on all performance metrics.

Table 6 presents the quadratic weight kappa scores of each model. The score of all models exceeds 0.86, indicating the consistency of the results with the original labels. Specifically, 384_B_16 achieves the highest quadratic kappa score (0.94), followed by 384_L_32 (0.935).

For 384_L_32, we plot its receiver operating characteristic (ROC), as illustrated in Figure 7. The model's AUC is 0.988.

This paper inputs the ViT fundus images and visualizes their attention maps to realize the operation of the attention mechanisms. Figure 8 illustrates that the attention mechanism helps the network to capture many valuable cues which are consistent with the fundus image labels.

TABLE 5 Four evaluation metrics for seven models

Models		Precision	Recall	Specificity	F1 score
224_B_16	0	0.828	0.934	0.936	0.878
	1	**0.941**	0.835	**0.984**	0.885
	2	0.877	0.862	0.96	0.869
	3	0.979	0.974	0.996	0.976
	4	0.99	0.983	0.999	0.986
224_B_32	0	0.803	0.95	0.923	0.87
	1	0.892	0.801	0.971	0.844
	2	0.851	0.766	0.956	0.806
	3	0.94	0.956	0.989	0.948
	4	0.975	0.977	0.996	0.976
224_L_32	0	0.82	0.925	0.933	0.869
	1	0.923	0.845	0.979	0.882
	2	0.865	0.833	0.957	0.849
	3	0.98	0.959	0.997	0.969
	4	0.984	0.984	0.998	0.984
384_B_16	0	0.827	0.966	0.936	**0.891**
	1	0.875	0.887	0.96	0.881
	2	**0.954**	0.778	**0.987**	0.857
	3	0.972	**0.979**	0.995	0.975
	4	0.98	0.991	0.997	0.985
384_B_32	0	0.815	0.958	0.931	0.881
	1	0.918	0.874	0.976	0.895
	2	0.925	0.805	0.978	0.861
	3	**0.99**	0.974	**0.998**	**0.982**
	4	0.96	**0.994**	0.994	0.977
384_L_16	0	0.792	**0.978**	0.919	0.875
	1	0.922	0.788	0.979	0.85
	2	0.863	0.782	0.958	0.821
	3	0.95	0.96	0.991	0.955
	4	0.972	0.974	0.996	0.973
384_L_32	0	**0.847**	0.893	**0.949**	0.869
	1	0.918	**0.898**	0.975	**0.908**
	2	0.897	**0.877**	0.966	**0.887**
	3	0.985	0.977	0.997	0.981
	4	**0.994**	0.985	**0.999**	**0.989**

Note: Bold metrics indicate the highest metrics in the same category. These values are mainly concentrated in the three models 384_B_16, 384_B_32, and 384_L_32.

4 | DISCUSSION

To the best of our knowledge, applying ViT for DR grade recognition has not been investigated yet. Thus, opposing the norm exploiting CNNs for DR grading recognition, in this work we evaluate seven ViT architectures on DR grade recognition. Thus, to highlight the performance capabilities of our ViT method, we challenge our ViT models against current CNNs exploiting the same

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TABLE 6 Quadratic kappa score

Model	Quadratic kappa score
224_B_16	0.934
224_B_32	0.913
224_L_32	0.923
384_B_16	0.94
384_B_32	0.933
384_L_16	0.92
384_L_32	0.935

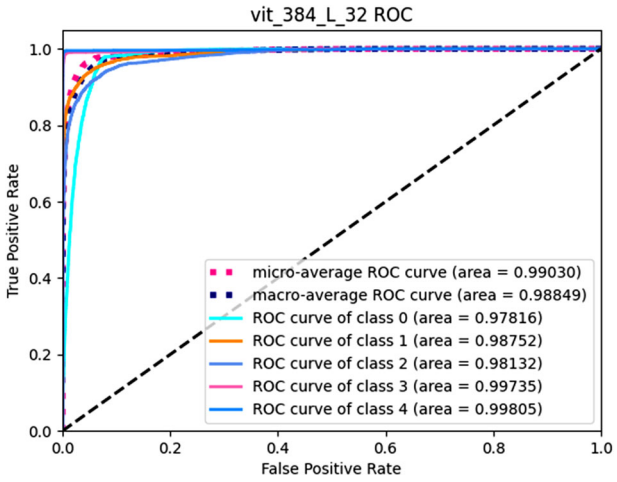


FIGURE 7 ROC of 384_L_32, demonstrates the classifier's performance under different thresholds. The macro-average is set as the final model's area under curve (AUC)

dataset, despite ViT and CNN being different in terms of architecture and feature extraction process.

Therefore, we train and test ResNet50 and ResNet101,[36] which are considered classic CNN models often used as a baseline, under the same experimental setup. Considering the two fundus image resolutions exploited, the corresponding notations are 224_ResNet_50, 224_ResNet_101, 384_ResNet_50, and 384_ResNet_101, respectively.

Table 7 illustrates that the ViT model test accuracy is higher than ResNet's. Considering the various resolutions and ResNet's depths, we observe that the ResNet model is regionally stable. However, ViT has a slight accuracy improvement considering the higher resolution and deeper models. Table 7 also presents the corresponding performance metrics of both models, highlighting that for the same resolution ViT manages a more appealing performance than ResNet.

A study has demonstrated that ResNet is effective in preventing adversarial attacks.[37] This paper compares the ability of two different models of 384_L_32 and 384_ResNet_101 to prevent adversarial attacks. Following the work of the literature,[37] we generate perturbations utilizing the fast gradient sign method (FGSM).[38] Different degrees of perturbations are added to the original image to form the adversarial samples, as shown in Figure 9. An attack is considered successful once the model misclassifies the adversarial sample. Table 8 illustrates the success rate of the two models attacked with different perturbation coefficients (ε), as the perturbation coefficients become larger, the change in the success rate of 384_ResNet_101 being attacked is

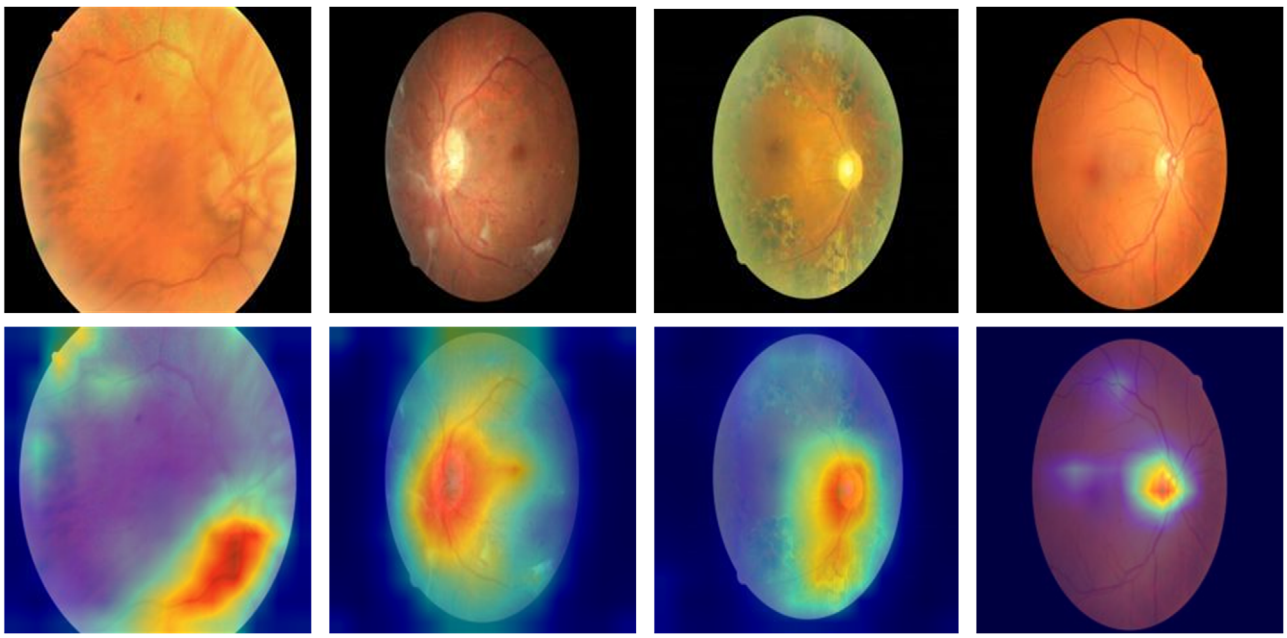


FIGURE 8 Performance of attentional mechanisms in fundus image recognition. In the upper row the raw fundus images and in the lower row the corresponding attention feature maps

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TABLE 7 Five evaluation metrics for six models

Model	% (95% CI) Precision	Sensitivity	Specificity	F1 score	Qks	Accuracy
224_ResNet_50	90.4 (82.9–97.9)	90.0 (80.8–99.1)	97.1 (93.6–100)	90.0 (83.9–96.0)	0.922	0.888
224_ResNet_101	90.5 (82.3–98.8)	90.1 (82.8–97.4)	97.1 (94.0–100)	90.2 (83.3–97.1)	0.923	0.889
224_B_16	**92.3 (83.7–100)**	**91.8 (83.5–100)**	**97.5 (94.2–100)**	**91.2 (84.8–99)**	**0.934**	**0.905**
384_ResNet_50	90.2 (80.2–100)	89.4 (78.7–100)	96.9 (93.0–100)	89.5 (81.5–97.6)	0.926	0.88
384_ResNet_101	90.8 (82.5–99.1)	89.7 (83.7–95.7)	97.1 (93.9–100)	90.1 (84.2–95.9)	0.928	0.889
384_L_32	**92.8 (85.2–100)**	**92.6 (86.3–98.9)**	**97.7 (95.1–100)**	**92.3 (85.9–99.5)**	**0.935**	**0.914**

Note: Bold values denote the ViT model, and each metric of ViT is higher than that of CNNs at the same resolution.
Abbreviations: CI, confidence interval; CNN, convolutional neural network; Qks, quadratic kappa score; ViT, Vision Transformer.

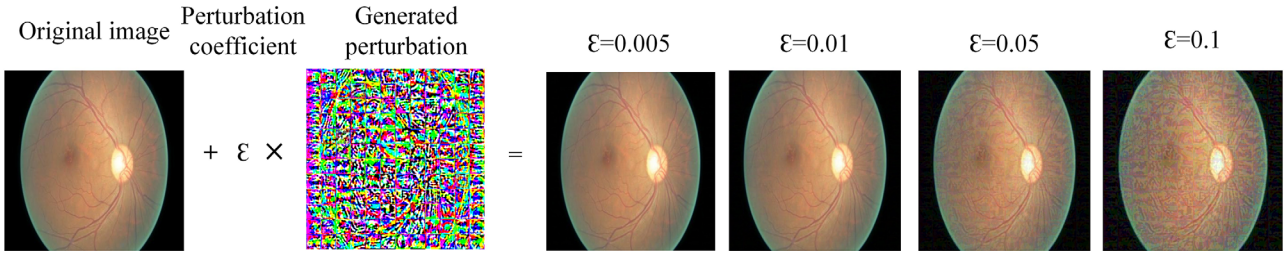


FIGURE 9 Adversarial samples with different perturbation coefficients. The changes in fundus images after adding perturbations are insignificant, but mapping the adversarial samples to the original sample labels is difficult for deep learning models. The generated perturbation may deceive the model and cause it to misclassify

TABLE 8 Attack success rate with different perturbation coefficients

Model	Attack success rate (%) 384_L_32	384_ResNet_101
$\varepsilon = 0$	0	0
$\varepsilon = 0.005$	0.28	1.11
$\varepsilon = 0.01$	0.60	2.25
$\varepsilon = 0.05$	3.80	13.85
$\varepsilon = 0.1$	7.98	31.86

Note: ε denotes the perturbation coefficient, and the lower the attack success rate, the more effective the model is in preventing adversarial attacks.

much greater than that of 384_L_32, thus illustrating the effectiveness of ViT preventing adversarial attacks than ResNet.

The ViT has several distinct advantages over CNN: (a) It can capture long-term dependencies between pixels. (b) It provides a built-in saliency that provides insight into what points the model is focusing on. (c) It is more effective in preventing adversarial attacks than CNNs. However, depending on the task, the depth of CNNs can scale from a few to hundreds of layers, which is a flexibility that ViT does not have.

Current works investigating DR grade recognition on the Diabetic Retinopathy Detection dataset are in Refs. 3 and 39–44 with Table 9 presenting their performance against our 384_L_32 model. The latter table highlights that our proposed 384_L_32 model is more appealing than current solutions in terms of SP,[42] Qks,[3] and AUC.[42,44] Due to the distribution imbalance between classes in the dataset, the models in Refs. 39–41 solely employ accuracy as the evaluation metric, which is not objective and comprehensive. Additionally, Van Grinsven et al.[44] employ multiple performance evaluation metrics, but this work exploits additional 53 576 unlabeled fundus images and does not propose a data annotation method. Nevertheless, lacking ground truth data does not afford objective performance evaluation. Additionally, the same work does not present any statistical analysis. In contrast, our model is comprehensively evaluated, showing competitive performance and indicating that our proposed ViT-based DR grade recognition scheme is feasible.

5 | CONCLUSIONS

Current DR grade recognition tasks rely on CNN and extreme learning machines, managing appealing results. Opposing the current trends, this paper employs a ViT for DR grade recognition. ViT relies on a powerful attention mechanism to capture the long-term pixels' dependencies, achieving appealing recognition performance and becoming a viable alternative to CNN. Our proposed 384_L_32 model attains 91.4% accuracy, 0.977 (95% CI (0.951–1)) specificity, 0.928 (95% CI (0.852–1)) precision, 0.926 (95% CI (0.863–0.989)) sensitivity, 0.988 AUC, and 0.935 quadratic

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TABLE 9 Performance comparison

Authors	Year	Approach	Dataset	Results
Roy et al.[3]	2017	CNN + dictionary	DRD	Qks = 0.86
Mansour et al.[39]	2018	AlexNet + LDA	DRD	Acc = 97.93%
Xu et al.[40]	2017	CNN	DRD	Acc = 94.5%
Wan et al.[41]	2018	VggNet	DRD	Acc = 95.68%
Zago et al.[42]	2020	CNN	DRD	AUC = 0.912 (95% CI (0.897–0.928))
				SP = 0.940 (95% CI (0.921–0.959))
Nazir et al.[43]	2019	Extreme learning machine	DRD	Acc = 99.6%
				Pre = 0.991
				Recall = 0.9932
				AUC = 0.995
Van Grinsven et al.[44]	2016	CNN	DRD	AUC = 0.972
384_L_32 (this paper)	-	Vision Transformer	DRD	Acc = 91.4%
				Pre = 0.928 (95% CI (0.852—1))
				Recall = 0.926 (95% CI (0.863–0.989))
				SP = 0.977 (95% CI (0.951—1))
				F1 score = 0.923-95% CI (0.859–0.995)
				Qks = 0.935
				AUC = 0.988

Abbreviations: Acc, accuracy; AUC, area under curve; CI, confidence interval; CNN, convolutional neural network; DRD, Diabetic Retinopathy Detection; LDA, linear discriminant analysis; Pre, precision; Qks, quadratic kappa score; SP, specificity.

weighted kappa score. We challenge the proposed method against current methods under the same experimental setup and demonstrate that our model is competitive, indicating that the attention mechanism-based ViT model is promising for the DR grade recognition task.

The ViT model has many more parameters than CNNs and must be pre-trained on large datasets and then transferred to specific tasks. Since the ViT model is relatively deep, exploiting training data in small batches is practical but imposes a more extended training and validation time (37 h for the 384_L_16 model against 2 h for a typical CNN). Additionally, due to the data imbalance, it is necessary to use additional technical means to balance the data set, such as a weighted loss function during training and weighted metrics during the model's evaluation. Future work shall investigate alternative methods for compressing transformers and generating lightweight models. Additionally, testing the performance of ViT under different numbers of layers and heads is also worth examining. Finally, this study has some potential limitations, such as the large dataset required for pre-training.

ACKNOWLEDGMENTS

This study is supported by the Key Projects of the National Natural Science Foundation of China (No. 81630104) and Major Project for Guang-Zhou Collaborative Innovation of Industry-University-Research (No. 201704020196).

The authors would like to express their gratitude to EditSprings (https://www.editsprings.cn/) for the expert linguistic services provided.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available at https://www.kaggle.com/c/diabetic-retinopathy-detection/data.[35]

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How to cite this article: Wu J, Hu R, Xiao Z, Chen J, Liu J. Vision Transformer-based recognition of diabetic retinopathy grade. *Med Phys*. 2021;48:7850–7863. https://doi.org/10.1002/mp.15312

APPENDIX A

Some suggestions on the use of the Vision Transformer (ViT) code

Given that deep learning (DL) requires researchers to have specialized domain knowledge, to accommodate researchers without a coding background, we offer some suggestions and instructions for using ViT.

To train DL networks, computer scientists typically set up local servers with high computational power or purchase expensive DL-ready workstations. An alternative is to purchase computational resources provided by cloud services to train DL models. Another cheaper way is to use Google Colaboratory. All calculations are performed in the cloud using Google Colaboratory, circumventing the need to purchase or install graphical processing units (GPUs) and associated software. Trained DL models can then be used directly in the cloud or downloaded and used locally.[45]

For ViT, we can download the model and pre-training parameters by clicking the link in the article. Then, load the model according to the following sample code:

```
from models import ViT
model = ViT(
  'B_16',
  # 'B_32',
  # 'L_32',
  # 'B_16_imagenet1k',
  # 'B_32_imagenet1k',
  # 'L_16_imagenet1k',
  # 'L_32_imagenet1k',
  pretrained = True,
  num_classes = 5 # Number of categories in the dataset
)
```

where 'B_16', 'B_32', 'L_32', 'B_16_imagenet1k', 'B_32_imagenet1k', 'L_16_imagenet1k', 'L_16_imagenet1k', and 'L_32_imagenet1k' are optional pre-trained models.

In case the researcher wants to try a different number of heads and layers, he can modify the parameters in the config.py file, as shown in the following code:

```
def get_base_config():
  return dict(
    dim = 768,
    ff_dim = 3072,
    num_heads = 12,
    num_layers = 12,
    attention_dropout_rate = 0.0,
    dropout_rate = 0.1,
    representation_size = 768,
    classifier = 'token'
  )
```

Note that the file needs to be saved after the parameters are modified. For model training, validation and testing, and preservation, the ViT model is the same as other deep learning models, and researchers can access the details by clicking on the links in the paper.

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